

Online Empirical and Methods Supplement

Public CAD Is Not Operational Ground Truth

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Scope

This supplement follows the main paper’s empirical sequence: source lineage and parsing; the July transition; initiation-stream and current-denominator evidence; the Ida daily path; the primary full-count comparison; then standardized, bootstrap, secondary-statistic, institutional, and reported-DV details. Extended identified-set proofs, constrained-program certificates, and formal-verification material are kept in the separate *Legacy Mathematical and Formal-Verification Archive* and are not appended to the professor-facing paper.

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1 Source lineage and parsing rules

The analysis distinguishes four evidence classes.

1. **First-hand public bytes:** official DataNOLA CSVs and their hashes.
2. **Deterministic aggregate transforms:** daily, half-day, monthly, and annual state counts derived from those files.
3. **Institutional documents:** official metadata, data dictionary, policy, oversight, and audit reports that define labels or exclude candidate explanations.
4. **Latent operational objects:** internal dispatch events, queue states, availability, entry histories, overwrite logs, and export transformations, which are not observed.

The first three classes identify a public-record regime change and delimit candidate interpretations. They do not turn the fourth class into observed data. Accordingly, J_{01} is always a public configuration—arrival field present, dispatch field absent—rather than a claim of physical arrival without dispatch.

1.1 Annual public-source lineage

1.1.1 Dataset binding

Table 1 binds each analysis year to the official DataNOLA identifier and the first-hand cache audit. Each annual bundle hash is computed by sorting the twelve monthly filenames and hashing the filename plus each file’s SHA-256 digest. This avoids concatenating or redistributing the raw source files. The monthly cache did not retain a reliable acquisition-date field, so the table reports that limitation rather than converting filesystem timestamps into retrieval dates. All sources were reverified on 25 August 2026.

Table 1. Annual DataNOLA source lineage, 2020–2024

Year	Dataset	Observed creation range	Rows	Files	Annual bundle SHA-256
2020	hp7u-i9hf	1 Jan.–31 Dec. 2020	432,892	12	a1d3d7fffb6a34e7...da7cc0db1
2021	3pha-hum9	1 Jan.–31 Dec. 2021	428,315	12	53b0ae39abbbc363...a7edc559
2022	nci8-thrr	1 Jan.–31 Dec. 2022	354,207	12	1ca797cd30109a80...e3d1e4d8
2023	pc5d-tvaw	1 Jan.–31 Dec. 2023	325,091	12	68d583fe0c4dbb43...4f011a13a
2024	2zcj-b6ts	1 Jan.–31 Dec. 2024	327,696	12	041f7ac52458c797...b4f6f464

Exact observed minimum and maximum timestamps, full hashes, monthly-file hashes, byte counts, local cache mtimes, official URLs, parser version, and verification date are in `r15_annual_source_lineage.csv` and `r15_public_source_audit.json`. Retrieval dates were not recorded in the monthly cache.

The annual row counts are raw public-file rows before the paper’s initiation restriction. Analysis denominators are constructed after parsing `SelfInitiated`. No narratives or addresses are used. Item identifiers are used only by the upstream canonical pipeline for deduplication and are not written to review artifacts.

1.1.2 Official metadata and changelog

The official 2021 page identifies OPCD as data provider, reports 21 public columns, supplies a data-dictionary attachment, and warns about preliminary classifications, later changes, mechanical or human error, sequencing, and over-time comparisons (City of New Orleans, 2021). When

inspected, its public “Dataset Changelog” stated that no changes had been archived. This is itself an observability limitation: the public page does not provide a dated change record that could explain 28 July 2021.

2 July transition: raw audit and aggregate parity

2.1 Parsing and state construction

The audit applies the same public-field definitions as the aggregate analysis:

$$D_i = \mathbf{1}\{\text{TimeDispatch}_i \text{ is nonblank}\}, \quad (1)$$

$$A_i = \mathbf{1}\{\text{TimeArrive}_i \text{ parses and is no earlier than TimeCreate}_i\}. \quad (2)$$

Timestamps are parsed as local ISO strings with a T or single-space separator. Empty strings and the textual values `null`, `none`, `nan`, and `nat` are missing sentinels. All nonblank dispatch and arrival values in the audit interval parse. Format-class counts show no switch that could produce the break mechanically.

Table 2. Complete aggregate audit, 25–31 July 2021

Date	Stream	N	D nonblank	D parsed	A nonblank	A valid	J_{00}	J_{10}	J_{01}	J_{11}
25 Jul.	Non-officer	721	631	631	532	532	90	99	0	532
26 Jul.	Non-officer	733	663	663	565	565	70	98	0	565
27 Jul.	Non-officer	731	658	658	585	585	73	73	0	585
28 Jul.	Non-officer	620	509	509	502	502	77	41	34	468
29 Jul.	Non-officer	666	510	510	532	532	87	47	69	463
30 Jul.	Non-officer	705	542	542	559	559	98	48	65	494
31 Jul.	Non-officer	725	584	584	572	572	96	57	45	527
25 Jul.	Officer	377	377	377	377	377	0	0	0	377
26 Jul.	Officer	404	404	404	404	404	0	0	0	404
27 Jul.	Officer	479	479	479	479	479	0	0	0	479
28 Jul.	Officer	575	230	230	575	575	0	0	345	230
29 Jul.	Officer	504	5	5	504	504	0	0	499	5
30 Jul.	Officer	548	3	3	548	548	0	0	545	3
31 Jul.	Officer	389	1	1	389	389	0	0	388	1

2.2 What the audit resolves

The audit resolves three concerns. The first J_{01} date is present in official source bytes. The change survives a parseability audit. And the abrupt officer-stream shift shows that the non-officer pattern is part of a broader public-record transition rather than an isolated fluctuation in one classified group.

The independent raw-to-aggregate comparison checks fourteen date-by-stream rows and all four public states. All 56 state cells match exactly. This establishes that the July result in the aggregate series is a lossless transform of the audited source counts.

The audit does not resolve the institutional cause. It has no event-entry history, user action, intermediate CAD object, export log, or versioned retention rule. The strongest supported wording is therefore “public-recording transition associated with initiation status,” not a named software or operational mechanism.

3 Ida path and maximum-change statistics

The daily file path begins with the intuitive quantities used in the main paper. The public missing-dispatch share—among non-officer records with a valid arrival field, the share whose public dispatch field is missing—rises from 5–12 percent in the preceding week to 43.4 percent on 31 August and 49.3 percent on 1 September. On landfall day, 29 August, the valid-arrival share instead falls to 43.8 percent while dispatch-field coverage remains 72.9 percent. The machine-readable daily series contains 22 dates from 22 August through 12 September 2021; it also records the separate one-day missing-dispatch excursion to 34.7 percent on 10 September. The half-day comparison uses ten twelve-hour event cells and the corresponding cells seven days earlier.

For the primary full-count design, let $\tilde{p}_{w,b,s}$ be the unconditional share in state s for half-day b of window w . The largest percentage-point change is

$$M^{\text{full}}(w) = \max_{b,s} |\tilde{p}_{w,b,s} - \tilde{p}_{w-7,b,s}|. \quad (3)$$

It uses every non-officer record in each cell. Ida’s value is 0.5320. The post-change comparison contains 151 ordinary reference windows for which both the event and baseline sides begin after 28 July 2021; its largest reference value is 0.3098. This reference set was defined after the July transition was identified and is therefore a descriptive post-hoc comparison.

For half-day bin t , state $j \in \{J_{01}, J_{10}, J_{11}\}$, and common stratum x , let p_{tjx}^E and p_{tjx}^B be event and baseline shares and let w_{tx}^B be the baseline weight over common support. The standardized contrast is

$$G_{tj} = \sum_{x \in \mathcal{X}_t^E \cap \mathcal{X}_t^B} w_{tx}^B (p_{tjx}^E - p_{tjx}^B). \quad (4)$$

The standardized amplitude is $M_{\max} = \max_{t,j} |G_{tj}|$. The event window begins 29 August 2021 and contains ten twelve-hour bins; the baseline is seven days earlier.

The standardized stage-era universe consists of 153 retained ordinary-week references beginning on or after 1 July 2021. Sixty-six eligible full-universe references occur entirely before 28 July, when J_{01} is structurally absent, and 64 start before the 1 July stage cutoff. These facts explain why a common formula does not imply a common measurement regime.

Two additional full-count reference sets are reported for transparency:

- the inherited stage-era membership set, which removes support selection but retains the existing membership rule; and
- a post-change set requiring both event and baseline sides to begin after 28 July, while retaining the frozen context exclusions.

The second is the post-change comparison used as the primary descriptive reference in the main paper.

Table 3. Ida common-support coverage and record counts

Bin	Event N	Baseline N	Event coverage	Baseline coverage
B1	276	244	0.928	0.754
B2	674	335	0.901	0.612
B3	206	291	0.748	0.505
B4	495	433	0.919	0.815
B5	295	290	0.766	0.838
B6	464	421	0.873	0.812
B7	311	268	0.846	0.802
B8	464	379	0.851	0.836
B9	240	255	0.829	0.812
B10	453	416	0.834	0.851

4 Secondary standardized estimator and common-support audit

The minimum coverages are 0.748 and 0.505; record-weighted coverages are 0.861 and 0.770. The symmetric 0.90 rule fails. Consequently, the standardized result cannot be described as the full event-window population. The full-count statistic uses all 3,878 event and 3,332 baseline records.

5 Bootstrap results and dependence caveats

5.1 Standardized windows

Within each observed stratum, records are resampled as multinomial draws over $(J_{00}, J_{10}, J_{01}, J_{11})$ while holding the observed stratum totals and baseline weights fixed. This is exactly the record-level nonparametric bootstrap available from the aggregate state counts. It captures record sampling conditional on the observed strata; it does not capture serial dependence, regime-boundary uncertainty, or design selection. Adjacent references share a week because one event period becomes the next baseline, so window-wise resampling does not preserve their cross-window dependence.

5.2 Full-count windows and rank distribution

The same multinomial logic is applied to unconditional half-day state counts. Ida’s full-count interval is $[0.4754, 0.6015]$. The strongest post-change reference point is 0.3098 with interval $[0.2599, 0.3799]$; the remaining top-five intervals have upper endpoints between 0.3636 and 0.3843. In each of 4,000 conditional window-wise draws, Ida is ranked against all 151 separately perturbed post-change references. Ida ranks first in every draw, but the exercise does not model temporal dependence and is not formal rank inference for an exchangeable time series.

An alternating-window sensitivity retains every second chronological post-change reference, which removes the direct event-to-next-baseline overlap. Ida is rank 1/77 including Ida in one phase and 1/76 in the other. The two largest retained reference values are 0.298 and 0.310. This does not eliminate longer-run dependence; it reports exactly how the result changes when adjacent overlap is removed.

Table 4. Bootstrap intervals for Ida and the five strongest standardized references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5072	0.4597	0.5656
15 May 2022	0.3282	0.2655	0.3954
22 May 2022	0.3083	0.2528	0.3725
12 May 2024	0.3064	0.2502	0.3880
5 May 2024	0.2904	0.2355	0.3660
10 Dec. 2023	0.2656	0.1630	0.3764

Table 5. Full-count bootstrap: Ida and five strongest post-change references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5320	0.4754	0.6015
15 May 2022	0.3098	0.2599	0.3799
10 Dec. 2023	0.2978	0.2140	0.3843
19 Jun. 2022	0.2959	0.2384	0.3636
5 May 2024	0.2941	0.2501	0.3735
12 May 2024	0.2926	0.2238	0.3773

6 Secondary statistics, LP, thresholds, and timing placebo

The 95th-percentile maximum-cell thresholds are 0.2094 in the full set, 0.2247 in the stage-era set, and 0.2057 in the same-season set. They yield 9, 8, and 9 exceeding cells. Reporting one threshold without the other two would overstate the stability of a binary label.

The constrained LP is a robustness calculation. Its Ida lower endpoint exceeds 0.25 by 0.000734, while 50 of 217 reference intervals have width above 0.01 and the maximum width is 0.070. The numerical LP status is therefore a tested numerical result, not an exact rational certificate. Formal code verifies arithmetic and finite certificate logic; it does not verify the empirical meaning of public fields.

The shifted-timing exercise is unfavorable to an institutional-timing interpretation: zero-padded translations of the support library fit better than the documented alignment. This does not alter the public-field contrast, but it means that the timing calculation does not identify a mechanism.

Table 6. Secondary stage-era statistics

Statistic	Ida	Largest reference	Rank including Ida
$M_{\max} / A$	0.5072	0.3282	1/154
σ_1	1.0837	0.6011	1/154
σ_2	0.2623	0.2451	1/154
σ_3	0.0858	0.1112	11/154
U_{direct}	0.4223	0.3282	1/154
$U_{\text{full upper}}$	0.2507	0.2125	1/154
Q_{upper}	0.4944	0.9408	100/154
D_{upper}	0.1716	0.2835	4/154

7 Excluded episodes and recovery comparisons

The frozen exclusion list omitted Tropical Storm Cristobal, which made Louisiana landfall on 7 June 2020 (National Hurricane Center, 2021). The 7 and 14 June windows therefore remain ordinary members of the full-qualified rank; the second uses the Cristobal week as its baseline. Their large pre-break full-count movements occur through the available J_{11}/J_{10} states, not through J_{01} . This post-hoc label documents an incomplete exclusion list and is not used to alter membership. The other labelled episodes in the table are excluded from the ordinary-week rank. The large post-Ida statistic compares a recovery week with Ida itself, while Francine is much smaller in the later regime. These comparisons reinforce the need to distinguish event stress from the measurement object used to observe it.

Table 7. Labelled emergency and disturbance windows: full-count maximum-cell contrast

Episode window	Event records	Baseline records	Raw $M_{\max}$
Hurricane Laura, 23 Aug. 2020	3,316	3,515	0.084
Hurricane Zeta, 25 Oct. 2020	3,963	3,638	0.129
Tropical Storm Cristobal, 7 Jun. 2020	3,431	3,323	0.333
Following comparison week, 14 Jun. 2020	3,454	3,431	0.340
February freeze, 14 Feb. 2021	2,851	3,329	0.113
Hurricane Ida, 29 Aug. 2021	3,878	3,332	0.532
Post-Ida week, 5 Sep. 2021	3,827	3,878	0.478
Hurricane Francine, 8 Sep. 2024	2,886	2,882	0.110

8 Initiation-stream and current-denominator evidence

The main analysis excludes officer-initiated records because their public dispatch convention differs radically. Their pattern is nevertheless analytically valuable. It shows that J_{01} is a normal public configuration in one stream after the break and that pooling initiation streams creates a denominator artifact. It does not prove that non-officer calls were operationally converted into officer activity.

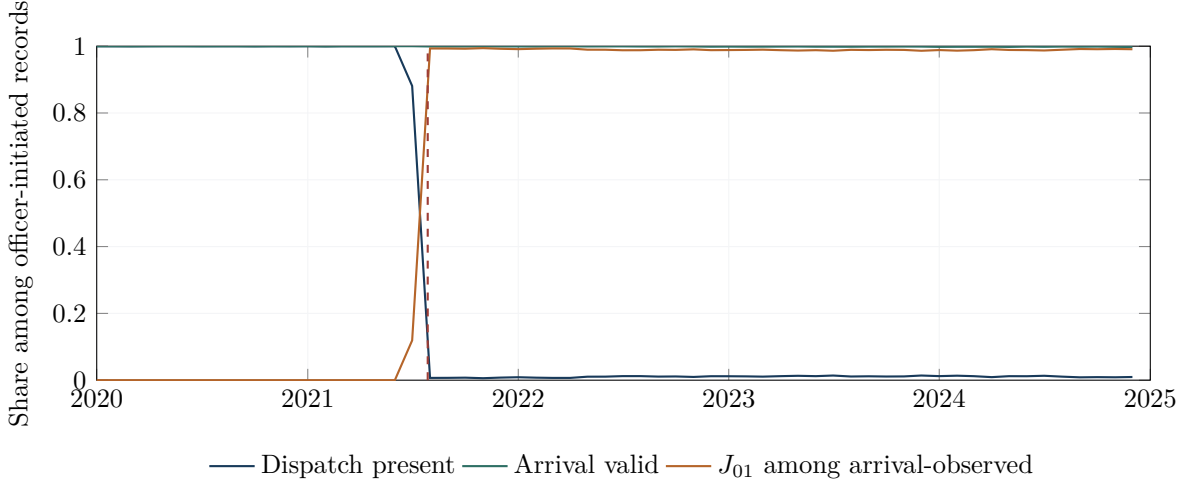


Figure 1. Officer-initiated public-field pattern, 2020–2024. Dispatch presence moves from essentially complete to approximately 1 percent while arrival remains essentially complete.

Table 8. Annual field completeness by initiation stream

Year	Stream	Records	Dispatch share	Arrival share
2020	Non-officer	271,868	0.903	0.801
2021	Non-officer	258,160	0.845	0.783
2022	Non-officer	243,179	0.745	0.724
2023	Non-officer	225,389	0.693	0.722
2024	Non-officer	212,314	0.654	0.742
2020	Officer	161,024	1.000	1.000
2021	Officer	170,155	0.617	1.000
2022	Officer	111,028	0.010	0.999
2023	Officer	99,702	0.012	0.999
2024	Officer	115,382	0.011	0.998

9 Disposition accounting

Among arrival-observed records, let p_{dt} be the recorded-disposition share and q_{dt} the dispatch-field rate within disposition d at time t . The aggregate change decomposes exactly as

$$\Delta \bar{q} = \sum_d \bar{p}_d \Delta q_d + \sum_d \bar{q}_d \Delta p_d, \quad (5)$$

where bars denote midpoints. For the two largest late Ida contrasts, the totals -0.4950 and -0.4824 decompose into within-disposition terms -0.5069 and -0.4946 plus composition terms $+0.0119$ and $+0.0123$. The observed final-disposition mix is not sufficient to explain the shift. Because the sample conditions on arrival observation and uses final recorded dispositions, the decomposition does not identify operational routing or causal mechanisms.

Table 9. 2025 and 2026 snapshot field completeness

Year	Denominator	Records	Dispatch present	Dispatch share
2025	Non-officer	207,050	136,712	0.6603
2025	Officer	122,720	1,117	0.0091
2025	All rows	329,770	137,829	0.4180
2026 snapshot	Non-officer	130,346	87,129	0.6684
2026 snapshot	Officer	79,483	738	0.0093
2026 snapshot	All rows	209,829	87,867	0.4188

9.1 Current-data denominator audit

The 2025 all-row rate is arithmetically correct but scientifically noncomparable to the non-officer series. The comparable series is 0.6538 (2024), 0.6603 (2025), and 0.6684 (2026 snapshot). The overall rate is low because officer-initiated records have nearly universal arrival fields and almost no dispatch fields under the later public convention.

10 Institutional-source audit

Table 10. Public institutional documents and what they identify

Source	Public status	What it adds	What it does not add
DataNOLA 2021 page and dictionary	Official page and attachment public	Field labels, provider, schema, preliminary-data warning, no archived changelog	Entry history, overwrite rule, internal/public lineage, July change cause
OIPM Hurricane Ida report	Official PDF public	Oversight context, emergency conditions, monitored activities	Machine-event semantics, export history, causal explanation of J states
NOPD manual chapters	Public editions located	Intended communications, call-response, and records practices	Complete exact-edition set for July–September 2021; proof of implementation
OPCD Carbyne APEX notice	Official page indexed; current URL returns not found	Cutover date of 24 June 2022	Explanation of July 2021 break
OIG Hexagon report	Official PDF public	OnCall Records project history; system was not launched	Evidence of a CAD-export transition

These documents supply genuine missing pieces: source semantics, public warnings, chronology, and negative evidence against two named technology stories. They still leave the decisive bridge missing. A mechanism claim would require a versioned changelog or internal/public reconciliation showing how a particular event and entry history became the released row.

11 Reported-DV measurement implications

The paper conditions on reported calls created in CAD. It does not observe latent need, attempts to seek help, contacts not converted into CAD items, or underlying domestic-violence incidence.

Broad text or code rules can also create false positives, and public suppression can create false negatives. The recommended classification remains $1/0/U$ until an authoritative crosswalk resolves the uncertain group.

If n_1 , n_0 , and n_U are the definitely included, definitely excluded, and unresolved counts, then the sharp no-assumption prevalence bounds are

$$\frac{n_1}{n_1 + n_0 + n_U} \leq p \leq \frac{n_1 + n_U}{n_1 + n_0 + n_U}. \quad (6)$$

For response-time outcomes, endpoint coverage must be stated on the full declared denominator. A duration among rows with both timestamps is a selected-stage statistic unless missing endpoints are bounded or independently qualified. The system-level Ida stress test shows why these requirements are substantive rather than cosmetic.

12 Reproduction and machine-consistency contract

The package build performs the following sequence:

1. rebuild aggregate diagnostics and all tables from the packaged frozen tally and reference artifacts;
2. optionally rerun the raw-source lineage audit against the canonical official cache;
3. compile the main paper, this supplement, research status note, and separate legacy archive;
4. combine only the main paper and empirical supplement into the professor-facing PDF;
5. check exact headline values, rank denominators, support status, threshold counts, 2025/2026 denominator arithmetic, source-audit invariants, and forbidden revision-history language in the main source;
6. verify PDF names, page relationships, text presence, manifest hashes, and the frozen reproduction snapshot.

The ordinary builder uses Python’s standard library and NumPy. SciPy is required only by the inherited constrained-optimization reproduction path, and pypdf is used to combine PDFs. Pandas is not required. The package manifest covers all curated files except the manifest itself and any generated zip. Text normalization performed by packaging is checked through the rebuilt numerical and PDF parity tests rather than treated as byte identity with an external working tree.

13 Interpretive checklist

Before using a public CAD duration as a performance measure, report:

1. the source identifier, observed date range, row count, and source hash;
2. the declared denominator and initiation-stream rule;
3. the exact start and endpoint semantics;
4. field nonblank, parseability, and ordering rates;
5. missing-endpoint handling and support coverage;
6. versioned institutional lineage, if available;
7. whether the target is a released-field statistic, selected-stage measure, bounded performance functional, or unavailable latent object;
8. shock exclusions, day-of-week, holidays, heat, weather, seasonality, COVID-era behavior, re-

porting behavior, help-seeking, suppression, and classification uncertainty.

References

- City of New Orleans (2021). Calls for service 2021. Data.NOLA dataset 3pha-hum9; official page, disclaimer, attached data dictionary, and empty archived changelog verified 25 August 2026.
- National Hurricane Center (2021). Tropical cyclone report: Tropical storm cristobal (al032020). Official report documents Louisiana landfall at 2200 UTC on 7 June 2020; verified 25 August 2026.

14 Numerical claim index

Table 11 identifies the package object that supplies each headline number. The release verifier recomputes the claims from these machine-readable objects; the table is an orientation aid, not an independent evidentiary source.

Table 11. Headline claims and their machine-readable sources

Claim	Reported value	Primary package object
Standardized Ida maximum	0.5072; rank 1/154	r15_aggregate_diagnostics.json
Full-count stage-era result	0.5320; rank 1/151 including Ida	r15_aggregate_diagnostics.json
Post-hoc post-change result	0.5320; rank 1/152 including Ida	r15_aggregate_diagnostics.json
Alternating non-overlap sensitivity	ranks 1/77 and 1/76 including Ida	r15_1_nonoverlap_sensitivity.csv
Threshold exceedances	9, 8, and 9	r15_aggregate_diagnostics.json
Standardized bootstrap interval	[0.4597, 0.5656]	r15_bootstrap_window_maxima.csv
Full-count bootstrap interval	[0.4754, 0.6015]	r15_bootstrap_window_maxima.csv
2025 non-officer dispatch share	$136,712/207,050 = 0.6603$	r15_current_denominator_audit.csv
2025 officer dispatch share	$1,117/122,720 = 0.0091$	r15_current_denominator_audit.csv
First non-officer J_{01} day	28 July 2021	r15_raw_july_25_31_audit.csv
Raw/aggregate July state parity	56/56 state cells match	r15_1_raw_aggregate_parity.csv
Annual public-source bindings	five official dataset IDs	r15_annual_source_lineage.csv

The aggregate objects are fully contained in the reproduction snapshot. The raw July and annual-lineage objects add first-hand source checks against the canonical public cache. Institutional documents remain a separate evidence class: they inform field semantics, warnings, and chronology, but do not generate or alter the numerical statistics above.